

Farzad Avishan

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Summary

I earned my Ph.D. in Industrial Engineering in 2023 from Ozyegin University and conducted postdoctoral research at HEC Montréal. I am currently a Research Scientist at Air Canada. My main research areas are combinatorial optimization and optimization under uncertainty, and I have explored problems across transportation, humanitarian, electric-vehicle, and production domains.

Current Position

Research Scientist, Data & AI — Air Canada, Canada

October 2024–Present

Main responsibilities:

Apply optimization techniques to support business decision-making.

Establish processes for large-scale data analysis, model development, validation, and implementation.

Conduct exploratory data analysis to identify trends, patterns, and relationships.

Collaborate with data engineers and machine-learning engineers to ensure data-pipeline and model integration.

Education

Postdoctoral Research Fellow — HEC Montréal, Canada

2023–2024

Supervisors: Jean-François Cordeau, Yossiri Adulyasak, and Okan Arslan

Ph.D. in Industrial Engineering - Ozyegin University, Turkey

March 2023

Supervisor: İhsan Yanıkoğlu

Thesis: *Applications of Robust Optimization in Logistics and Production Planning* (CGPA: 3.87/4.00)

M.Sc. in Industrial Engineering - Mazandaran University of Science and Technology, Iran

September 2016

Supervisor: Iraj Mahdavi

Thesis: *A New Optimization Model for Production and Distribution of Supply Chain with Green Logistic Approach*

B.Sc. in Industrial Engineering - Tabriz University, Iran

September 2014

Published Papers

Farzad Avishan, Amira Dems, Yossiri Adulyasak, Okan Arslan, and Jean-François Cordeau (2026). "Inventory Routing with Heterogeneous Vehicles and Hazardous Material Backhauling." *Transportation Research Part E: Logistics and Transportation Review*, 205, 104491.

Farzad Avishan, Mehmet Berk Karasu, Melike Çap, and İhsan Yanıkoğlu (2025). "Optimization of Drone Base Station Locations and Mobile Charging Drone Routing for Post-Disaster Communication." *Computers & Operations Research*, 107206.

Farzad Avishan, İhsan Yanıkoğlu, and Mehmet Soysal (2025). "Adaptive Optimization Approach for Production and Distribution Planning of Perishable Food Products under Demand Uncertainty." *Annals of Operations Research*, 1–48.

Farzad Avishan, İhsan Yanıkoğlu, and Yaseen Alwesabi (2023). "An Optimization Approach for Electric Bus Scheduling under Travel Time and Energy Consumption Uncertainty." *Transportation Research Part C*.

Farzad Avishan, Milad Elyasi, İhsan Yanıkoğlu, Ali Ekici, and Okan Örsan Özener (2023). "Humanitarian Relief Distribution Problem: An Adjustable Robust Optimization Approach." *Transportation Science*.

Yaseen Alwesabi, **Farzad Avishan**, İhsan Yanıkoğlu, Zhaocai Liu, and Yong Wang (2022). "Robust Strategic Planning of Dynamic Wireless Charging Infrastructure for Electric Buses." *Applied Energy*, 307, 118243.

Ongoing Research

Farzad Avishan, İhsan Yanıkoğlu, and Okan Arslan. “Safe Reformulations of Quadratic and Block Separable Bilinear Uncertainty: Method and Applications.” Target journal: *Operations Research*.

Farzad Avishan, Yossiri Adulyasak, Okan Arslan, and Jean-François Cordeau. “A Multi-Period Optimization Framework for Mobile Charging Station Relocation Problem.” Target journal: *Management Science*.

Farzad Avishan, Yossiri Adulyasak, Okan Arslan, and Jean-François Cordeau. “Delivery and Charging Scheduling Planning for Last-Mile Delivery with Electric Vehicles.”

Farzad Avishan. “An Inventory Routing Problem for Relief Distribution under Demand Uncertainty.”

Teaching Experience

Optimization Under Uncertainty — Teaching Assistant and Co-instructor

Heuristic Optimization Methods — Teaching Assistant and Co-instructor

Introduction to Industrial Engineering — Teaching Assistant

Simulation Modelling and Analysis — Teaching Assistant

Engineering Economics — Teaching Assistant

Selected Presentations

Farzad Avishan, İhsan Yanıkoğlu, and Yaseen Alwesabi. “An Optimization Approach for Electric Bus Scheduling under Travel Time and Energy Consumption Uncertainty.” ISMP Conference, July 2024, Montreal, Canada.

Farzad Avishan, Amira Dems, Yossiri Adulyasak, Okan Arslan, and Jean-François Cordeau. “Inventory Routing Problem with Heterogeneous Vehicles: Delivery and Backhauling.” Optimization Days, 2024, Montreal, Canada.

Farzad Avishan, Milad Elyasi, İhsan Yanıkoğlu, Ali Ekici, and Okan Örsan Özener. “Humanitarian Relief Distribution Problem: An Adjustable Robust Optimization Approach.” European Humanitarian Operations (EURO HOPE) Conference, October 2022, Istanbul, Turkey.

Hani Shahmoradi-Moghadam, Jörn Schönberger, **Farzad Avishan**, and İhsan Yanıkoğlu. “An Adjustable Decentralized Robust Optimization Approach for Mobile Supply Chains with Shared Factories.” 31st European Conference on Operational Research, July 2021, University of West Attica, Athens, Greece.

İhsan Yanıkoğlu and **Farzad Avishan**. “Robust Reformulation of Ambiguous Chance Constraints with Discrete Probability Distributions.” 30th European Conference on Operational Research, June 2019, Dublin, Ireland.

Farzad Avishan, Iraj Mahdavi, and Hassan Rezazadeh. “Developing a New Mathematical Model for Production and Distribution of Supply Chain with Green Logistic Approach.” 12th International Conference on Industrial Engineering (ICIE 2016), January 25–26, 2016, Kharazmi University, Tehran, Iran.

Academic and Industrial Projects

PI: İhsan Yanıkoğlu; co-PIs: **Farzad Avishan** and Berk Karasu. “Utilizing Historical Data to Develop a Predictive Model for Forecasting Transportation Costs in Upcoming Periods.” Funded by Clear D3, March–September 2023.

PI: İhsan Yanıkoğlu; co-PIs: Basak Altan and Mehmet Soysal. “Applications of Optimization under Uncertainty Methods on Production Planning, Supply Chain and Logistics Problems.” Funded by TUBITAK-3501, February 2022–February 2024.

PI: Okan Örsan Özener; co-PIs: Basak Altan and Ali Ekici. “Cooperative Game Theory Approaches for Collaborative Dynamic Lot Sizing Problem.” Funded by TUBITAK-1001, November 2018–November 2021.

Technical Skills, Interests, and Expertise

Combinatorial optimization

Design and management of large-scale networks

Heuristic algorithms

Machine learning

Generative AI: LangChain, LangGraph, and AI agents

Software Skills

Programming languages: Python, C++

Optimization and analytical software: Gurobi, FICO Xpress, CPLEX, GAMS, MATLAB

Version control: Git, GitHub

Journal Refereeing

Transportation Science

Transportation Research Parts B, C, and E

Omega

PLOS ONE

Annals of Operations Research

Computers & Industrial Engineering

International Journal of Production Research

References

Prof. İhsan Yanıkoğlu — Ozyegin University; Ph.D. advisor; ihsan.yanikoglu@ozyegin.edu.tr

Prof. Okan Arslan — HEC Montréal; postdoctoral advisor; okan.arslan@hec.ca

Prof. Yossiri Adulyasak — HEC Montréal, Canada; postdoctoral advisor; yossiri.adulyasak@hec.ca

Prof. Jean-François Cordeau — HEC Montréal, Canada; postdoctoral advisor; jean-francois.cordeau@hec.ca